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Fish biomass and functional diversity recovery following a fishing ban in Luoma Lake of the Huai River Basin, China

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Abstract

Background Fish functional response to conservation measures such as fishing bans is very important for lake biodiversity conservation but remains less studied.

Methods We conducted a 6-year fish community investigation in Luoma Lake of the Huai River Basin in China, including three years (2018–2020) before and three years (2021–2023) after a full fishing ban (i.e., complete removal of fishing pressure). Taxonomic and functional diversity indices, including species richness, Shannon and Simpson diversity index, Pielou's evenness, functional richness (FRic), functional evenness (FEve), and functional dispersion (FDis), were used to assess community-level changes. Community-weighted mean (CWM) values were calculated from 12 functional traits, and R-mode linked to Q-mode (RLQ) analysis combined with the fourth-corner test was conducted to link traits to environmental gradients.

Results We found no significant change in fish abundance and taxonomic diversity, but fish biomass and functional richness increased significantly after the fishing ban. Non-metric multidimensional scaling (NMDS) and similarity percentage (SIMPER) analysis indicated that the fish community underwent significant compositional changes following the fishing ban. Trait–environment analysis showed that age at first maturity, mouth position, body shape, feeding habit, and life history strategy were significantly associated with environmental gradients.

Conclusions These results suggest that the fishing ban promoted early functional recovery and ecological reorganization processes of the fish community, highlighting the functional mechanisms underlying assemblage restructuring.

Keywords Fishing ban, Freshwater ecosystem, Functional diversity, Trait-based analysis, Luoma Lake

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Introduction

Freshwater ecosystems support high levels of biodiversity, including about 9.5% of all known animal species and one-third of all vertebrates (Balian et al. 2008). However, freshwater biodiversity is currently in crisis (Harrison et al. 2018), with freshwater fishes identified as the most threatened vertebrate group after amphibians (Darwall and Freyhof 2015). According to the latest Living Planet Report, monitored populations of freshwater taxa (including fish and other species) have declined by an average of 85% between 1970 and 2020, representing the most severe decline among all wildlife groups (WWF

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2024). This dramatic loss is driven by intensifying human activities, including overfishing, habitat destruction, and water pollution (Vörösmarty et al. 2010; Acreman et al. 2020). Such pressures lead to ecosystem degradation and biodiversity loss, underscoring the urgent need for effective conservation strategies. While well-managed protected areas have proven effective in safeguarding terrestrial biodiversity (Gaston et al. 2008), freshwater ecosystems continue to face both persistent and emerging threats (Dudgeon et al. 2006; Reid et al. 2019). In this context, fishing bans have emerged as a critical management tool to mitigate over exploitation and support ecological recovery in aquatic systems (Hilborn et al. 2020). Assessing the effectiveness of such conservation, however, requires robust indicators that can capture not only changes in species numbers but also shifts in ecological functions and community organization.

Traditional taxonomic diversity metrics provide a fundamental baseline for quantifying species richness and composition, and are widely used for assessing biodiversity patterns across spatial and temporal scales (Clarke and Warwick 1998). They are particularly valuable for detecting broad-scale changes in community composition and for enabling comparisons across studies and long-term monitoring programs, while also providing a standardized and widely comparable framework for assessing biodiversity change across different ecosystems and time periods. However, such indices may fail to capture the nuanced functional dynamics of community responses to disturbance, whereas trait-based approaches provide deeper insights into the ecological mechanisms underlying community reassembly (Mouillot et al. 2013). In particular, trait-based approaches can detect changes in fish communities earlier than traditional taxonomic methods, providing a more sensitive means of assessing ecological shifts (Xia et al. 2025). For instance, the proliferation of opportunistic species may mask the decline of equilibrium-strategy or top-trophic-level species, ultimately compromising long term ecosystem stability (Winemiller 2005). Therefore, integrating taxonomic diversity metrics with functional diversity indices is critical to comprehensively assess the multidimensional responses of fish communities following fishing bans. Relying solely on traditional taxonomic approaches, such as species richness or abundance, may overlook critical ecological shifts, as these metrics cannot fully capture changes in species interactions, resource utilization strategies, or ecosystem roles (Mason et al. 2005). Functional homogenization in fish assemblages can occur even when taxonomic indices suggest stable species numbers, because the dominance of opportunistic species with similar ecological niches reduces functional differentiation within communities (Mouchet et al.

2010). This pattern has been well documented in fish systems, where functional homogenization can exceed changes observed at the taxonomic level (Villéger et al. 2013, 2014). Similar mechanisms have also been reported in other ecosystems, where taxonomic stability obscures deeper structural homogenization within communities, as shown in plant assemblages by Keith et al. (2009). Functional diversity analyses, which evaluate traits such as feeding habits, body size, reproductive strategies, and habitat preferences, provide mechanistic insights into how communities reorganize ecologically. The combined use of taxonomic and functional diversity metrics enables researchers to discern whether recovery trends reflect true ecological restoration (Spake et al. 2015; Villéger et al. 2017; Rypel 2023). Such combined taxonomic and functional approaches are essential for evaluating ecological responses to fishing bans and informing adaptive management strategies, thereby supporting the long-term recovery and conservation of aquatic ecosystems and the ecological services they provide.

In addition to anthropogenic pressures such as fishing, environmental gradients are widely recognized as fundamental drivers of community assembly in freshwater ecosystems. Variables such as nutrient concentrations, dissolved oxygen, temperature, and habitat structure act as ecological filters that constrain species distributions and select for specific functional traits (Tonn et al. 1990; Jeppesen et al. 2000; Jackson et al. 2001). These environmental filters operate by favoring species with trait combinations suited to local conditions, thereby shaping both taxonomic composition and functional structure of communities. Importantly, strong anthropogenic disturbances, such as intensive fishing, can override or obscure these environmentally driven patterns by selectively removing particular species or trait groups (Shin et al. 2005; Edeline et al. 2007). As a result, disentangling the relative contributions of environmental filtering and fishing pressure is essential for understanding the mechanisms underlying community reassembly following management interventions. Integrative analytical approaches, such as RLQ (R-mode linked to Q-mode) analysis combined with the fourth-corner method, provide a powerful framework to explicitly link environmental variables, species composition, and functional traits, thereby allowing a more mechanistic assessment of community responses to both natural and anthropogenic drivers.

To address the depletion of inland fishery resources, fishing bans and seasonal moratoria have been widely implemented worldwide as key regulatory measures to promote the recovery of aquatic ecosystems. However, the effectiveness of these policies has often been compromised following the resumption of fishing activities, as economic incentives and the prevalence of illegal

practices have led to renewed and intensified exploitation (Chen et al. 2020; Liu et al. 2022). Furthermore, the outcomes of such bans have varied globally, with studies reporting positive (Du et al. 2025; Liu et al. 2023a, b; Mak et al. 2021; Watson et al. 2021; Xie et al. 2022), neutral (Bergman et al. 2015; Feng et al. 2023), and even negative ecological responses (Haase et al. 2025; Xu et al. 2022; Yuan et al. 2024; Zhou et al. 2025). Such variability reflects the complex interplay of biological, environmental, and anthropogenic factors, making it challenging to disentangle the direct effects of fishing bans from other co-occurring drivers. In addition, longer-term abundance and biomass information, which are critical for understanding population recovery and ecosystem functioning, are often lacking or analyzed separately from taxonomic and functional metrics, reducing the power to infer ecological processes. These limitations highlight the need for integrative approaches that combine taxonomic, functional, and population-level metrics to better evaluate the ecological effectiveness of fishing bans.

In recent years, China has launched its most comprehensive inland fishery conservation initiative to date, the ten-year full fishing ban in the Yangtze River Basin, aimed at facilitating large-scale ecological recovery (Chen et al. 2020; Mei et al. 2020; Xiong et al. 2026). As the policy has been progressively expanded, its geographic coverage has extended beyond the Yangtze River Basin to include other associated major river and lake systems. Luoma Lake, situated within the Huai River Basin, represents a key example of this policy's spatial extension. The lake has historically supported intensive commercial fisheries, with fishing activities primarily conducted using pelagic trawls, gillnets, and trap nets, targeting economically important species such as *Coilia nasus*, *Cyprinus carpio*, *Carassius auratus*, *Neosalanx tangkahkeii*, and *Hypophthalmichthys nobilis*, among others, which has led to a severe depletion of fishery resources. On May 10, 2021, a full fishing ban was officially enacted in Luoma Lake to support the restoration of aquatic biodiversity and the improvement of overall ecosystem health (<http://nynct.jiangsu.gov.cn>). Despite the increasing implementation of such policies, empirical evaluations of their ecological effects in non-Yangtze systems remain limited. To address this gap, the present study integrates taxonomic indices, abundance and biomass patterns, and multi-trait functional diversity across six consecutive years in Luoma Lake, providing a comprehensive assessment of fish community reassembly before and after the fishing ban.

In this study, we applied taxonomic diversity indices, functional diversity metrics, community-weighted mean and R-mode linked to Q-mode (RLQ) combined with the fourth-corner test, to address the following objectives: (1)

to evaluate the impacts of the fishing ban on fish community structure and taxonomic and functional diversity; (2) to identify response patterns of key functional traits and their linkages to environmental gradients; and (3) to evaluate whether observed changes in community structure, functional diversity, and functional traits are consistent with ecological recovery and functional reassembly following the fishing ban. The findings are intended to inform the refinement of fishery management policies and contribute to the development of more comprehensive strategies for freshwater ecosystem restoration.

Materials and methods

Study area and monitoring plan

A 6-year study was conducted from 2018 to 2023 in Luoma Lake (34°00′–34°14′N, 118°05′–118°19′E), located in northern Jiangsu Province, China. Luoma Lake spans the cities of Xuzhou and Suqian, with an average water depth of 3.3 m and a surface area of approximately 296 km². It is the fourth largest freshwater lake in Jiangsu Province and the third largest in the Huai River Basin. Luoma Lake functions as a flow-through system within the Yi-Shu-Si River Basin of the Huai River network. It is hydrologically connected to the Beijing–Hangzhou Grand Canal via the Zhongyun Canal. The lake primarily receives inflow from the Yihe River and the Zhongyun Canal. Outflow is regulated through three main channels: the Zhangshan Sluice, which discharges into the Xinyi River; the Zaohe Sluice, which discharges into the Zhongyun Canal; and the Yanghetan Sluice, which discharges into the Liutang River (Yan et al. 2017; Xing et al. 2018). On May 10, 2021, a ten-year full fishing ban was officially implemented in the lake, prohibiting all commercial fishing activities. In addition, recreational angling in Luoma Lake is strictly limited to “one person, one rod, one line, and one hook”, which keeps any subsistence or recreational harvest at negligible levels. Based on the timing of this regulatory intervention, we monitored the lake for three years before the ban (2018–2020) and three years after its implementation (2021–2023).

Fish sampling

A total of 14 sampling sites were established across Luoma Lake to ensure a comprehensive spatial coverage of the lake area (Fig. 1). From 2018 to 2023, fish surveys were conducted annually during the autumn season (September to November). At each site, one floating gillnet, one sinking gillnet, and one trap net were deployed. The multi-mesh gillnet used in this study was adapted from the Swedish standardized design (Appelberg et al. 1995). Each gillnet consisted of 12 mesh panels with knot-to-knot sizes ranging from 5 to 55 mm (Guo et al. 2020). The trap net consisted of

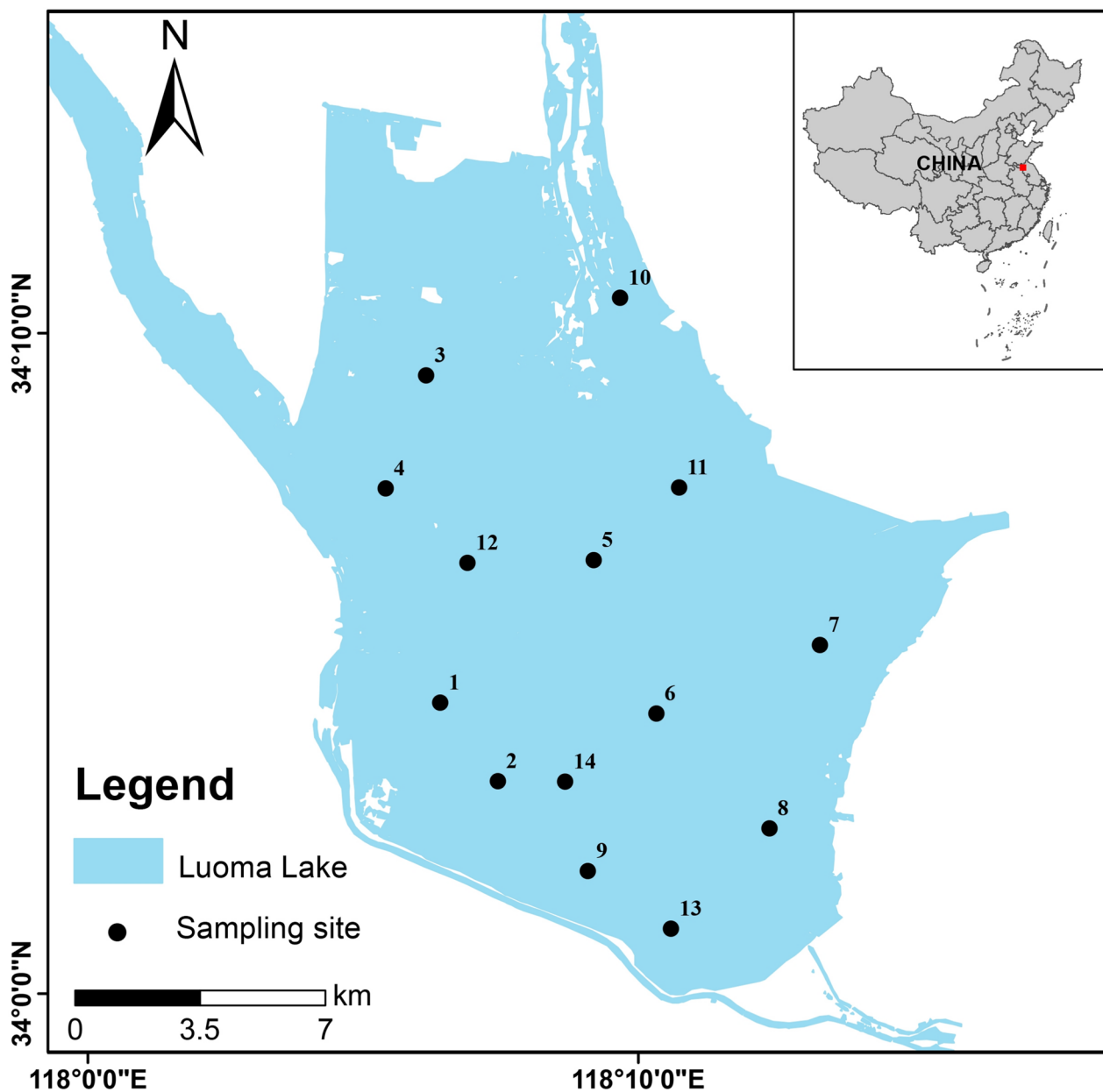


Fig. 1 Study area and fish sampling sites in Luoma Lake of the Huai River Basin, China

5-mm mesh and was 25 m long, divided into 20 compartments, with two 10-cm funnel openings located at the front and rear to allow fish entry. The multi-mesh gillnets and trap net were set at 18:00 on the first day and retrieved at 06:00 the following morning, resulting in a fishing duration of 12 hours. All captured fish were identified to species, counted, measured for length, and weighed. More details about the sampling gears and procedures were presented in previous studies (Guo et al. 2020; Liu et al. 2023a, b). To ensure data comparability, the types of fishing gears and operation

protocols were kept consistent during the whole monitoring period. Simultaneously, water temperature, pH, dissolved oxygen (DO), and electrical conductivity (EC) were measured in situ using a YSI Professional Plus multiparameter water quality meter (YSI Inc., Yellow Springs, OH, USA). Water depth was recorded with a portable depth sounder. In addition, water samples were collected from a depth of 0.5 m below the surface using a 5 L plexiglass water sampler. At each site, 2 L of water was collected and stored in 1 L polyethylene bottles, and transported with ice for laboratory analysis.

Water chemical parameters, including total nitrogen (TN) and total phosphorus (TP), were determined according to standard analytical methods (APHA 2005; Xiong et al. 2022; Gao et al. 2023).

Taxonomic diversity

To assess the taxonomic diversity of fish communities, four commonly used indices were calculated: Species Richness, Shannon Index, Simpson Diversity Index, and Pielou Evenness Index. Species Richness refers to the total number of distinct species present in a given area or ecosystem, representing the most basic measure of biodiversity (Harrison and Cornell 2008). Shannon Index quantifies community diversity by incorporating both species abundance and evenness (Shannon and Weaver 1949). It is calculated based on the proportional abundance of each species and increases with both the number of species and the evenness of their distribution. Simpson Diversity Index quantifies community diversity based on the probability that two individuals randomly selected from a sample belong to different species (Simpson 1949). Higher values indicate greater diversity. Compared with the Shannon Index, the Simpson Diversity Index is more sensitive to dominant species and less influenced by rare species. Pielou Evenness Index evaluates the evenness of species distribution within a community (Ricotta and Avena 2003). It is derived by standardizing the Shannon Index against the maximum possible value (i.e., when all species are equally abundant), and ranges from 0 (uneven) to 1 (completely even). The taxonomic diversity indices of the fish were calculated using the "vegan" (Oksanen et al. 2022) package in the R4.3.3 (R Core Team 2024).

Functional traits

A total of 12 functional traits were compiled to characterize fish ecological strategies, including six continuous traits: vulnerability, length at first maturity, growth rate, lifespan, age at first maturity, and trophic level, and six categorical traits: feeding habit, life history strategy, depth range, mouth position, body shape, and population resilience. These traits were selected to represent five functional axes proposed by Winemiller et al. (2015) and Villéger et al. (2017): resource acquisition, locomotion, life history, predator avoidance, and trophic balance. Life-history strategy was treated as a categorical trait, and species were assigned to the equilibrium, opportunistic, or periodic categories according to published classifications, primarily following Liu et al. (2022). Detailed trait information was obtained from published literature, books, monographs, and

the FishBase database (www.fishbase.org) and given in Table S1 (Yang et al. 2024).

Functional diversity

To quantify the functional diversity of fish assemblages, we employed three core indices: Functional Richness (FRic), Functional Evenness (FEve), and Functional Dispersion (FDis) (Villéger et al. 2008). FRic reflects the amount of functional trait space occupied by the species within a community. It describes the extent of ecological niche utilization and indicates the range of functional strategies present. FEve measures the regularity of species distribution in the functional trait space, as well as the evenness of their abundances. It is computed using the Minimum Spanning Tree (MST) approach, which requires at least three species per sampling unit to ensure valid estimation. FDis describes the degree to which the relative abundances of species are distributed toward the extremes of the functional trait space. It is calculated based on the deviation of each species from the community centroid, weighted by species abundance. All functional diversity indices were calculated using the "FD" package (Laliberté and Legendre 2010) in R version 4.3.3, based on the same set of 12 functional traits described above.

Community-weighted mean of trait values

The Community-Weighted Mean (CWM) of trait values was used to quantify the central tendency of functional trait distribution within fish communities by integrating species-level trait values weighted by their relative abundances or biomass (Grime 1974; Lavorel et al. 2008). For continuous traits, CWMs were computed as weighted means based on species abundance or biomass. For categorical traits, traits were transformed into binary variables, and CWMs were calculated as the relative contribution of each trait category to total abundance or biomass. This index reflects the dominant functional characteristics of the community, and all CWM calculations were based on the same set of 12 functional traits described above to ensure consistency with the functional diversity analyses.

Statistical analysis

In this study, the Shapiro–Wilk test was first applied to assess the normality of data distributions. Based on the results, either a paired *t*-test or Wilcoxon test was used to compare differences in fish abundance, biomass, taxonomic diversity, functional diversity, and community-weighted means of individual traits between the before- and after-fishing ban periods.

To evaluate changes in fish community structure, non-metric multidimensional scaling (NMDS) was performed

based on Bray–Curtis dissimilarities using both species-level abundance and biomass data, which were analyzed separately, with values log-transformed to reduce the influence of dominant species (Mak et al. 2021). Permutational multivariate analysis of variance (PERMANOVA) was conducted separately for abundance and biomass data to test for significant differences in community composition between the two periods (Anderson 2001). When PERMANOVA results were significant ($p < 0.05$), the Similarity Percentage (SIMPER) analysis was used for each dataset to identify the species contributing cumulatively $\geq 70\%$ to the observed dissimilarity.

In addition, we performed R-mode linked to Q-mode (RLQ) analysis (Dolédec et al. 1996) combined with the fourth-corner method (Legendre et al. 1997) to examine relationships among environmental variables, species traits, and community composition, following the analytical framework described by Dray et al. (2014). Briefly, RLQ analysis integrates three data matrices, namely the site-by-species abundance matrix (L), the site-by-environment matrix (R), and the species-by-traits matrix (Q), and identifies synthetic ordination axes that maximize the covariance between environmental variables and functional traits. These synthetic ordination axes (RLQ axes) represent joint gradients summarizing the covariation between environmental conditions and species functional traits. The strength of this joint structure is quantified by eigenvalues and associated inertia, where projected inertia represents the proportion of covariance explained along each RLQ axis. The fourth-corner method was used to test the statistical relationships between individual functional traits and environmental variables by linking species traits to site conditions through species abundance data. The significance of trait–environment associations was assessed using 9999 permutations under model 6 (Dray and Legendre 2008). Model 6 combines the permutation of sites (model 2) and species (model 4) and provides the most robust control of type I error among the available permutation procedures. *P*-values were adjusted for multiple testing using Holm's correction. Within the RLQ framework, a global permutation test evaluates whether the overall association between environmental variables and functional traits is stronger than expected under a null model of random association among the L, R, and Q matrices. In addition, Model A and Model B tests assess whether species distributions are structured by environmental variables (Model A) or by functional traits (Model B), respectively. This approach allowed us to investigate whether environmental changes associated with the fishing ban were linked to shifts in functional trait composition, thereby helping to determine whether community reassembly was driven by environmental filtering or other ecological

processes. Statistical analyses were performed in R version 4.3.3, primarily using the packages "vegan" and "ade4" (Dray and Dufour 2007).

Results

Species composition and community structure

A total of 34 fish species were recorded in Luoma Lake from 2018 to 2023, belonging to 7 orders and 11 families (Table S2). Among these, Cypriniformes was the most dominant order, with 26 species, accounting for 76.47% of the total species richness. This was followed by Belontiiformes and Gobiiformes, each with 2 species (5.88%), while Clupeiformes, Hemiramphiformes, Siluriformes, and Anabantiformes were each represented by a single species (2.94%). During the period before the fishing ban, 27 species from 6 orders and 9 families were collected, whereas 32 species from 7 orders and 11 families were recorded after the ban. The dominant species before the ban were *C. nasus*, *Toxabramis swinhonis*, and *Pseudorasbora parva*, while the community after the ban was mainly composed of *C. nasus*, *T. swinhonis*, *C. auratus*, *Pseudobrama simoni*, and *N. tangkahkeii*. In terms of life-history strategies, 16 species were classified as opportunistic strategists, 11 as periodic strategists, and 7 as equilibrium strategists. Prior to the fishing ban, opportunistic species accounted for 84.84% of the total abundance, equilibrium species for 14.16%, and periodic species for 1.00%. After the ban, opportunistic species increased to 93.61%, while equilibrium and periodic species accounted for 0.91% and 5.47%, respectively. Compared with the period before the fishing ban, fish abundance in Luoma Lake showed no significant change after the ban. The average abundance per sampling site was 107.1 ± 74.83 individuals before the ban and 95.9 ± 73.59 individuals after the ban (Wilcoxon test, $p = 0.47$, Fig. 2a). In contrast, fish biomass increased significantly, from 1002.98 ± 684.73 g per site before the ban to 2145.99 ± 1558.14 g after the ban (Wilcoxon test, $p < 0.01$, Fig. 2b).

The NMDS ordination based on species abundance revealed a clear separation between samples from the periods before and after the fishing ban (Fig. 3a). This visual pattern was supported by the PERMANOVA, which indicated significant differences in community composition between periods ($R^2 = 0.105$, $F = 7.951$, $p = 0.001$; 999 permutations). The SIMPER analysis further showed that the top five contributing species accounted for approximately 72.0% of the total Bray–Curtis dissimilarity (Table S3). *C. nasus* was the largest contributor, accounting for 36.6% of the cumulative dissimilarity, although its abundance did not differ significantly between periods (54.47 vs. 46.30 individuals per site, $p = 0.512$). *P. parva* declined markedly after

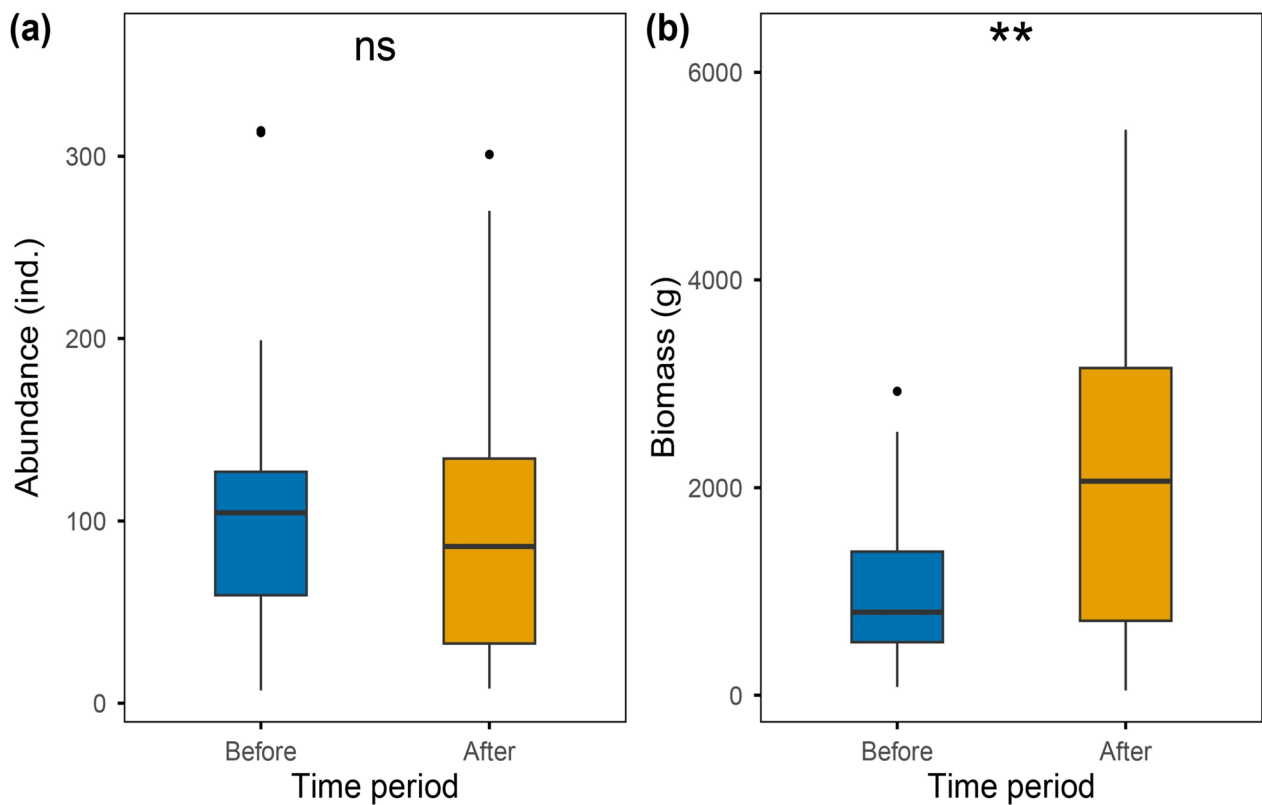


Fig. 2 Comparison of fish **a** abundance and **b** biomass before (2018–2020) and after (2021–2023) the fishing ban in Luoma Lake. Boxplots show the median, interquartile range (IQR), and outliers across all sampling sites. Asterisks indicate statistical significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$; "ns" indicates no significant difference

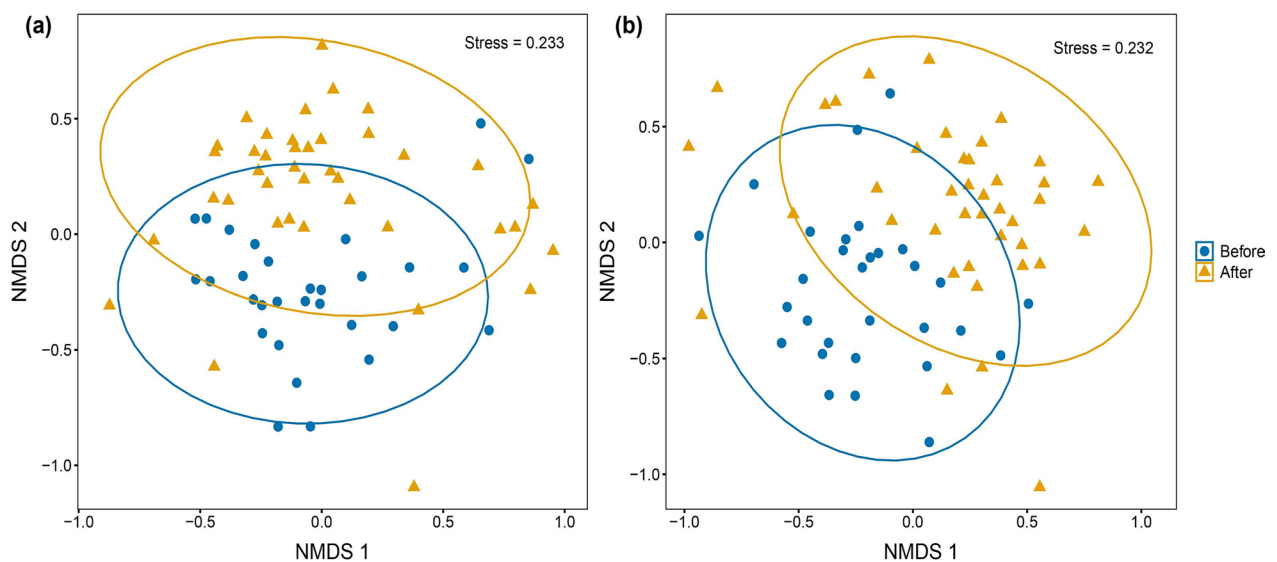


Fig. 3 Non-metric multidimensional scaling (NMDS) ordination of fish community composition based on Bray–Curtis dissimilarities for **a** abundance and **b** biomass. Ellipses indicate 95% confidence intervals for the periods before (blue) and after (orange) the fishing ban

the fishing ban (14.57 vs. 0.10 individuals per site, $p = 0.001$), whereas *N. tangkahkeii* increased in abundance (6.03 vs. 10.00 individuals per site, $p = 0.987$). *Protosalanx hyalocranius* also showed a significant decline from 6.40 to 0.78 individuals per site ($p = 0.001$). Together, these species were major contributors to the observed shift in community composition. Similarly, the biomass-based NMDS ordination also showed distinct clustering between the two periods (Fig. 3b). The corresponding PERMANOVA confirmed significant differences in biomass composition ($R^2 = 0.117$, $F = 9.000$, $p = 0.001$; 999 permutations), with the period effect explaining 11.7% of the observed variation. SIMPER analysis identified several species as major contributors to biomass dissimilarity between periods (Table S4). *C. nasus* contributed the largest proportion of cumulative dissimilarity (23.9%), although its biomass increase was not statistically significant (418.7 vs. 609.3 g per site, $p = 0.262$). Significant biomass increases were observed for *C. auratus* (98.5 vs. 315.5 g per site, $p = 0.046$) and *P. simoni* (10.4 vs. 207.6 g per site, $p = 0.044$), whereas *P. parva* declined markedly in biomass (80.7 vs. 0.6 g per site, $p = 0.008$).

Taxonomic and functional diversity

Following the implementation of the fishing ban, species richness, Shannon index, Simpson Diversity Index, and Pielou evenness index increased by 19.08%, 19.10%, 5.88%, and 3.66%, respectively, although none of these changes were statistically significant (Fig. 4a–d). Among the functional diversity metrics, FRic increased significantly by 45.24% after the fishing ban (Wilcoxon test, $W = 334.5$, $p = 0.009$; Fig. 4e). By contrast, FEve ($t = 0.185$, $p = 0.854$; Fig. 4f) and FDis (Wilcoxon test, $W = 506$, $p = 0.700$; Fig. 4g) showed no significant differences between periods.

Fish community functional traits

For abundance-weighted CWM, several categorical and continuous traits exhibited significant shifts following the ban (Fig. S1a). Specifically, the relative representation of species with inferior and subterminal mouth positions, as well as those with compressed body shapes and low population resilience, significantly increased after the fishing ban ($p < 0.05$). In terms of continuous traits, length at first maturity also showed a significant increase ($p < 0.05$). Conversely, the proportion of species associated with an equilibrium life-history strategy, those with superior mouth positions, and fusiform body shapes significantly declined after the ban ($p < 0.05$). Biomass-weighted CWM revealed partially consistent yet distinct patterns (Fig. S1b). Traits associated with benthic habitat preference, inferior and subterminal mouth positions, and the continuous traits length at first maturity, lifespan,

and age at first maturity were significantly higher after the fishing ban ($p < 0.05$), indicating a greater biomass contribution from species with longer life spans and delayed maturation. In contrast, the CWM values of equilibrium life-history strategists, demersal and pelagic habitat users, terminal mouth position, and growth rate significantly declined ($p < 0.05$).

Trait–environment relationships based on RLQ and fourth-corner analyses

The RLQ analysis revealed non-random co-structures among the environmental variables, fish community composition, and functional traits (Table 1). The first two RLQ axes explained 87.9% of the total variance, with eigenvalues of 0.718 and 0.279, respectively. Axis 1 accounted for 63.29% of the projected inertia, while Axis 2 explained 24.59%. The global RLQ test based on 9999 permutations indicated that the overall co-structure among matrices was non-random. However, the separate significance tests for Model A (species–environment associations) and Model B (species–trait associations) did not reach the 0.05 threshold.

In the RLQ ordination, sampling sites showed clear spatial separation along the first two axes (Fig. 5a). Environmental variables displayed distinct correlations with the ordination space, with total nitrogen (TN), total phosphorus (TP), temperature, conductivity, and dissolved oxygen (DO) loading positively on Axis 1, while pH and depth range were oriented in the opposite direction (Fig. 5b). The fourth-corner test further indicated that several fish functional traits were significantly associated with the RLQ axes (Fig. 5c). Specifically, age at first maturity, omnivorous feeding habit, equilibrium life-history strategy, superior mouth position, and fusiform body shape were positively associated with Axis 1, whereas terminal mouth position was negatively associated with this axis. Compressed body shape was positively associated with Axis 2, while growth rate was negatively associated with Axis 2. These relationships were further confirmed in the RLQ biplot, where the significant traits clustered along distinct directions of the ordination axes (Fig. 5d). Collectively, these findings indicate that variation in environmental conditions corresponded to specific combinations of functional traits within the fish assemblages, despite the non-significance of the global trait–environment tests, suggesting that trait–environment relationships may be driven by specific associations rather than consistent global patterns.

Discussion

Changes in community structure and functional traits

In the present study, NMDS and PERMANOVA analyses revealed significant differences in fish community

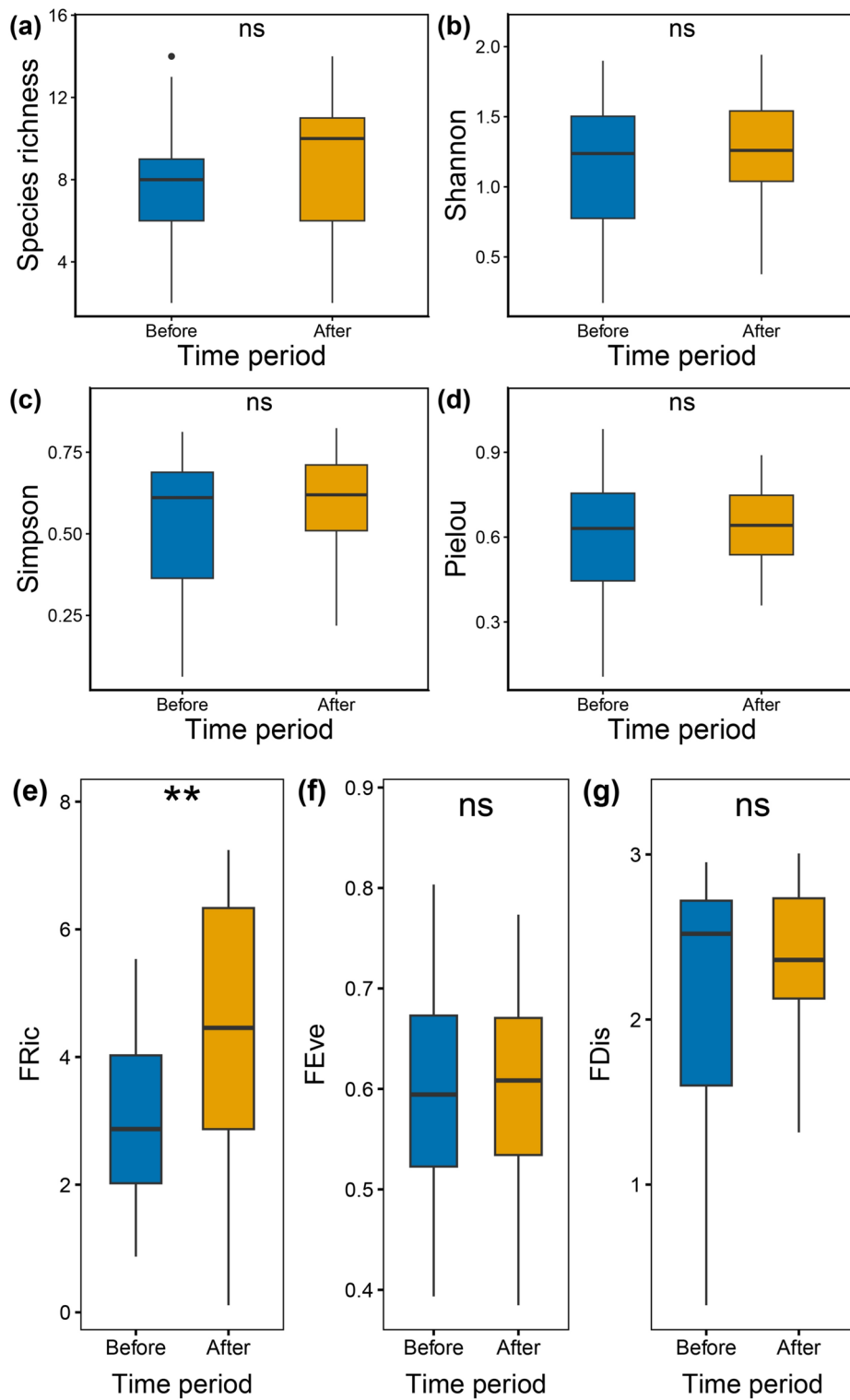


Fig. 4 Changes in taxonomic and functional diversity of fish communities before and after the fishing ban. Boxplots show comparisons in the periods before (blue) and after (orange) the fishing ban: **a** species richness, **b** Shannon diversity index, **c** Simpson diversity index, **d** Pielou's evenness, **e** functional richness (FRic), **f** functional evenness (FEve), and **g** functional dispersion (FDis). Asterisks indicate statistical significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$; "ns" indicates no significant difference

Table 1 Results of the RLQ analysis. The first two axes of RLQ analysis were extracted, and the global significance test was repeated 9999 times

RLQ analysis	Axis 1	Axis 2
Eigenvalues	0.718	0.279
Covariance	0.847	0.528
Correlation	0.361	0.202
Projected inertia (%)	63.29	24.59
Model A	$p > 0.05$	
Model B	$p > 0.05$	

Model A tested the relationship between species occurrence and environmental factors

Model B tested the relationship between species occurrence and fish functional traits. The null hypothesis, which assumed no relationship between fish functional traits and environmental factors, was rejected when the P values of both models were below the significance level of 0.05.

composition between the periods before and after the fishing ban, a pattern consistent with previous findings for lake fishes (Xie et al. 2022) and macroinvertebrate communities (Yang et al. 2023) in the Yangtze River. A range of studies have demonstrated that overfishing can cause sharp declines in long-lived, low-fecundity species (Botsford et al. 1997; Walsh et al. 2006). Conversely, fishing bans and moratoriums can promote the recovery of aquatic communities by reducing harvest-related disturbances and alleviating pressure on exploited species (Pipitone et al. 2023; Wang et al. 2021; Xie et al. 2022). In our dataset, fish abundance remained relatively stable, whereas biomass increased significantly after the ban. This indicates that although the number of individuals did not change markedly, the community shifted toward larger and heavier fish, consistent with the observed

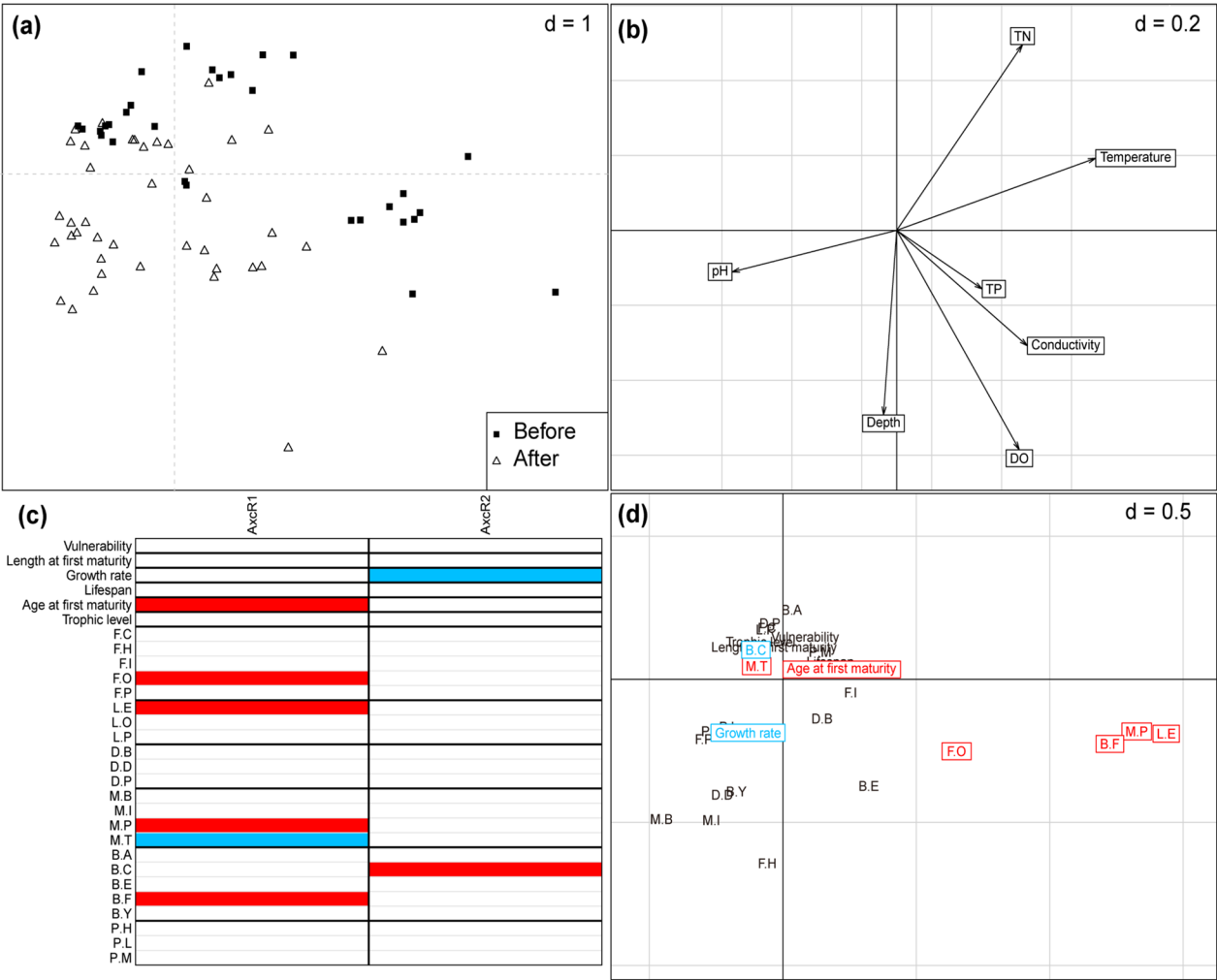


Fig. 5 **a** Ordination of sampling sites on the first two RLQ axes. **b** Correlations between environmental variables and RLQ axes represented by arrows. **c** Fourth-corner test showing the associations between fish functional traits and RLQ axes (D2 statistics). Red cells indicate significant positive associations, and blue cells indicate significant negative associations at $\alpha = 0.05$ after Benjamini–Hochberg correction. **d** Biplot of significant trait–axis associations projected in the RLQ ordination space, with red labels indicating positive and blue labels indicating negative associations

increases in length at first maturity. Such patterns are typical of systems where fishing pressure on large-bodied species has been relaxed (Carlson et al. 2007; Halpern and Warner 2002; Babcock et al. 2010).

The RLQ and fourth-corner analyses further revealed that community and trait reorganization were associated with environmental gradients, but these relationships should be interpreted within the context of reduced fishing pressure. These gradients likely reflect underlying habitat quality and productivity differences that act as environmental filters on fish functional traits (Jeppesen et al. 2000). Importantly, the fishing ban is unlikely to be the primary driver of environmental variation, but may have modified how fish communities respond to existing gradients. Under high fishing pressure, trait composition is often dominated by species with fast life-history strategies and small body sizes, which can obscure or weaken trait–environment relationships (Shin et al. 2005; Edeline et al. 2007). Following the reduction in fishing pressure, the partial recovery of larger-bodied and late-maturing species likely enhanced the responsiveness of community structure to environmental filtering, thereby strengthening the linkage between traits and environmental gradients. For instance, nutrient enrichment and oxygen availability are known to influence metabolic constraints and trophic strategies, favoring omnivorous species and individuals with flexible feeding habits (Pörtner and Farrell 2008; Killen et al. 2010). Similarly, habitat-related variables such as water depth and pH can regulate species distributions by altering habitat heterogeneity and physiological tolerances (Tonn et al. 1990; Jackson et al. 2001). These mechanisms are consistent with the observed shifts in traits such as age at first maturity, body shape, and feeding guilds. Nevertheless, the lack of significance in the global fourth-corner tests suggests that trait–environment relationships were not uniform across all species and traits. Instead, the observed patterns were likely driven by specific trait–environment associations rather than a consistent overall response, a phenomenon commonly reported in trait-based community studies (Dray et al. 2014; Li et al. 2021). This indicates that environmental filtering and fishing pressure jointly shape community assembly, with environmental effects becoming more detectable as anthropogenic disturbance is reduced. This pattern is consistent with the expectation that anthropogenic disturbance can mask environmental filtering, and that its relaxation allows niche-based processes to become more prominent in structuring communities.

Despite the pronounced changes in functional composition, alpha diversity indices remained relatively stable, suggesting that the overall richness and evenness of species were not strongly altered by the ban, a pattern consistent with previous observations in freshwater systems

(Liu et al. 2021). Instead, the main responses occurred at the trait level. These findings indicate a partial release from fishing-induced pressure. However, the observed shifts in functional traits, particularly the increase in opportunistic strategists, suggest that the community may currently be in a transitional recovery stage rather than having reached a stable state. The reassembly process is therefore likely ongoing, and longer-term monitoring would be required to determine whether the community will shift toward trait combinations characteristic of a more stable and less disturbed state.

Life-history strategies and functional reorganization

Fish species with distinct life-history strategies show differential sensitivities to fishing pressure (Wiedmann et al. 2014). Opportunistic strategists are typically characterized by short lifespans, high reproductive rates, and rapid resource exploitation (Meador and Brown 2015). Following the implementation of the fishing ban, reduced fishing pressure and increased resource availability likely enabled opportunistic species to respond rapidly, allowing them to expand their populations and remain dominant within the assemblage. In this study, opportunistic species accounted for 93.61% of total abundance after the ban. By contrast, equilibrium and periodic strategists generally exhibit longer generation times and lower fecundity, which may result in slower population recovery following reductions in fishing pressure. Similar patterns have been reported in other lake ecosystems, including Liangzi Lake (Feng et al. 2023).

Although overall abundance showed no significant change, the significant increase in biomass suggests a shift in community size structure following the fishing ban. This pattern may reflect a partial recovery of larger-bodied and slower-growing species, as widely observed after reductions in fishing pressure (Halpern and Warner 2002; Babcock et al. 2010). This interpretation is supported by the observed increases in community-weighted mean length at first maturity. Such patterns suggest that previous fishing activities selectively removed large-bodied species, contributing to community-level miniaturization, a well-documented consequence of size-selective harvesting (Law 2000; Sharpe and Hendry 2009). By contrast, both FDis and FEve remained stable, suggesting that although the functional trait space expanded, the internal distribution and dispersion of traits within the assemblage did not undergo substantial reorganization (Mason et al. 2005; Laliberté and Legendre 2010). These patterns indicate that functional recovery has begun but remains at an early or transitional stage, consistent with previous findings that functional structure often recovers more slowly than other community attributes (Mouillot et al. 2013).

Recovery status and future implications

In summary, the results indicate that the fish community is undergoing partial recovery following the fishing ban. The relaxation of fishing pressure facilitated a shift toward larger-bodied species and traits indicative of slower life-history strategies, such as increased length at first maturity (Jennings et al. 1999; Law 2000). However, the persistent dominance of opportunistic species, the unchanged alpha diversity indices, and the limited responses of functional evenness and functional dispersion suggest that the ecosystem has not yet reached a stable or functionally saturated state (Villéger et al. 2008; Mouillot et al. 2013). Instead, the fish community remains in an ongoing phase of restructuring (Scheffer et al. 2001). Additionally, the stability of key traits such as the vulnerability index and trophic level throughout the study period reinforces the view that the community has not fully transitioned to a mature or functionally saturated condition (Pauly et al. 1998; Devictor et al. 2010).

From a management perspective, these findings emphasize that the fishing ban can initiate recovery but may not be sufficient to achieve a functionally saturated and resilient community state within a limited time frame (Hilborn et al. 2003). Continued monitoring is essential for tracking changes in functional diversity and community structure. Management strategies such as restoring lakeshore zones, and preventing nutrient enrichment could further enhance population recovery (Smith et al. 1999). Integrating taxonomic and functional approaches, as demonstrated in this study, provides a comprehensive framework for evaluating the ecological effectiveness of fishing bans and guiding adaptive management in large lake ecosystems (Mason et al. 2005; Cadotte et al. 2011).

Conclusions

Our 6-year monitoring demonstrates that a full fishing ban can trigger measurable but uneven recovery in lake fish communities. Although overall abundance and conventional taxonomic diversity metrics showed limited responses, both fish biomass and FRic increased significantly after the ban, indicating a shift toward larger-bodied and functionally diversified species. Functional trait analyses further revealed clear links between trait composition and environmental gradients, suggesting that recovery is shaped jointly by the release from fishing pressure and habitat conditions.

Together, these findings indicate that recovery following a fishing ban represents an early stage of functional rebuilding rather than full ecological restoration. The partial improvement, combined with the persistence of opportunistic strategists and limited changes in several functional indices, highlights the need for continued

protection and complementary habitat management to reinforce resilience. By integrating taxonomic and functional perspectives, this study provides a robust framework for evaluating conservation outcomes and supports the role of fishing bans as an important tool for sustaining lake ecosystems.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13717-026-00726-0>.

Supplementary Material 1.

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Author contributions

YC designed the study. XZ, ZS, XQ, BY, HL, and WX collected the data. XZ analyzed the data. XZ and YC wrote the original draft of the manuscript. ZS, XQ, BY, HL, and WX reviewed and edited the manuscript. All authors read and approved the final manuscript.

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Data availability

The data sets used in this study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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